Cover Letter — Round 2 Revision

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Title: Numerical Spectral Correspondence between a Non-Autonomous Quadratic Map and the Riemann Zeros: An Exploratory Study
Journal: Mathematical and Computational Applications (MDPI)
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Dear Editor,

Thank you for the opportunity to revise our manuscript. We are grateful to both reviewers for their constructive and detailed feedback. This round of revision has substantially improved the clarity, scope, and epistemic transparency of the work.

Below we respond to each comment point-by-point, indicating where changes were made in the revised manuscript and, where appropriate, why certain suggestions were not adopted.

Summary of Major Changes

Before addressing individual comments, we summarize the principal revisions:

1. **New standalone Conclusions section** (§6) — now explicitly separates "Positive findings" from "Limitations" and makes the out-of-sample prediction failure and non-GUE local statistics **principal conclusions** rather than buried caveats.

2. **Manuscript shortened** — detailed parameter scans (Module A alternatives, Module B alternatives, microscopic ε-scan procedure) moved to three new appendix subsections; main text now focuses on the core map, low-order fit, OOS validation, and unfolded-spacing test.

3. **Terminology precision** — replaced ∼19 instances of vague/overstated language:

   - "ground-state zero" → "first non-trivial zero" / "lowest positive ordinate"
   - "phase transition" → "sharp numerical convergence transition"
   - "topological mutation" → "abrupt, discontinuous jump"
   - "intrinsic physical noise floor" → "baseline numerical residual" / "residual dispersion floor"
   - "rigorous higher-order perturbative form" → "additional empirical correction term"

4. **Single-author language** (Reviewer 2, Comments 1-2) — replaced all "we/our" with "the author/this work" throughout.

5. **Methods before Results** (Reviewer 2, Comment 5) — §3 Methods now precedes §4 Results.

6. **All abbreviations defined** (Reviewer 2, Comment 7) — verified on first appearance.

The revised manuscript is 35 pages (down from the original length due to appendix relocation), compiles cleanly with zero undefined references, and all 43 cited references have been verified as accurate and accessible.

Response to Reviewer 1

Main Comment 1: What is fitted, and what is predicted?

Concern: Several ingredients are calibrated using the same zeros against which the spectrum is evaluated. The out-of-sample test shows test MSE 1-2 orders larger than training MSE, and k_1 drifts with calibration window M. This should be a **principal conclusion**, not a technical limitation.

Response: Fully adopted.

- **New §6.2 "Limitations"** (lines 1302-1351) now lists as the **first principal limitation**: "The model does not predict unseen zeros" — with full OOS quantification (held-out MSE 1-2 orders larger, k_1 drifts $5.79 \to 9.15 \to 10.04$ across $M \in \{50, 70, 80\}$).
- Abstract (lines 40-48) now explicitly states the model is "an in-sample numerical correspondence, not an independent prediction" and quantifies the OOS gap.
- The closing summary (lines 1345-1351) restates: "the proposed system gives a phenomenological, calibrated reconstruction of part of the smooth counting-function behavior of the Riemann zeros, but it does not predict unseen zeros..."

On the cooling law: We now explicitly state (§2.2, lines 197-256) that the $1/\log^2 n$ term combines the known Riemann-von Mangoldt mean spacing with an *assumed* square-root eigenphase response (not derived from first principles), and the $1/\log^3 n$ term is "an additional empirical correction term" (line 228, revised from "rigorous higher-order perturbative form").

On k_1 drift: We do not interpret this as RG flow; the revised text (§4.1.1, lines 980-1029) presents it as optimizer sensitivity and multimodal fitting landscape, with no claim of a beta function or fixed point.

Main Comment 2: What operator is actually being diagonalized?

Concern: For non-autonomous systems, evolution is $P_{T-1} \cdots P_0$, not the time-averaged $\bar{P}_T$. The two spectra need not be related. Can the author compare these in a small system, or give an argument?

Response: We chose the **"give an argument"** branch of the reviewer's "or" condition.

- **§3.2 Module B** (lines 363-407) now contains a substantially strengthened epistemic disclaimer:
  - Explicitly states: "the spectrum of the ordered product $P_{T-1} \cdots P_0$ is unrelated to the spectrum of the arithmetic mean $\bar{P}_T$ in general, since averaging discards temporal ordering and non-commutative effects"
  - Candidly admits: "The author does not provide an adiabatic theorem, ergodic argument, or convergence proof"
  - Positions $\bar{P}_T$ as a "computationally tractable heuristic surrogate" justified only by slow variation of λ_n relative to mixing time

We did **not** perform the small-system numerical comparison because:

1. The reviewer offered "compare in a small system **or** give an argument" — we believe the revised argument satisfies the transparency requirement

2. A small-system ordered-product calculation would face the same numerical instabilities (underflow, phase accumulation) that motivated the time-averaged construction in the first place

3. If the reviewer or editor deems the argument insufficient, we are prepared to add this as a follow-up appendix in a subsequent minor revision

On sensitivity: Appendix G.4 "Additional Robustness Checks" (lines 1623-1726) now includes:

- Table of Module C eigenvalue-filtering choices (imag cutoff, branch selection, unwrapping tolerance)
- Quantified impact on final k_1 and MSE
- Seed robustness study (10 runs at fixed $M = 70$, CV $\approx 4.7\%$)

On Gaussian vs Monte Carlo consistency: We now explicitly state (lines 368-370, Methods) that the Monte Carlo large-N construction is "a related but distinct numerical construction" sharing the same underlying map but differing in spatial discretization, and we do not claim these are numerically identical in overlap regions.

Main Comment 3: GUE and the meaning of the large-N agreement

Concern: Globally rescaled GUE comparison concerns mean counting function (not unfolded local statistics). The unfolded test shows model eigenphases are Poisson, not GUE — this should be stated directly in Conclusions.

Response: Fully adopted.

- **New §6.2 "Limitations"**, item (2) (lines 1311-1325): "The model's own eigenphase spacings are not GUE" — now a principal conclusion, with full KS statistics quoted:
  - True Riemann zeros: $D(\text{GUE}) = 0.067$, $p = 0.74$; $D(\text{Poisson}) = 0.348$, $p < 10^{-4}$ *OK* GUE
  - Model eigenphases: $D(\text{GUE}) = 0.300$, $p < 10^{-4}$; $D(\text{Poisson}) = 0.082$, $p = 0.49$ *OK* Poisson
- Closing summary (lines 1345-1351) restates: "... but it does not... reproduce their GUE local statistics"
- §5.3 Discussion (lines 1212-1268) now clearly distinguishes:
  - "Mean counting-function agreement" (what the calibrated model achieves)
  - "Unfolded local spacing statistics" (where it fails — Montgomery-Odlyzko correspondence)

We agree the large-N agreement is not evidence against RMT or for a more fundamental description — this is now explicit.